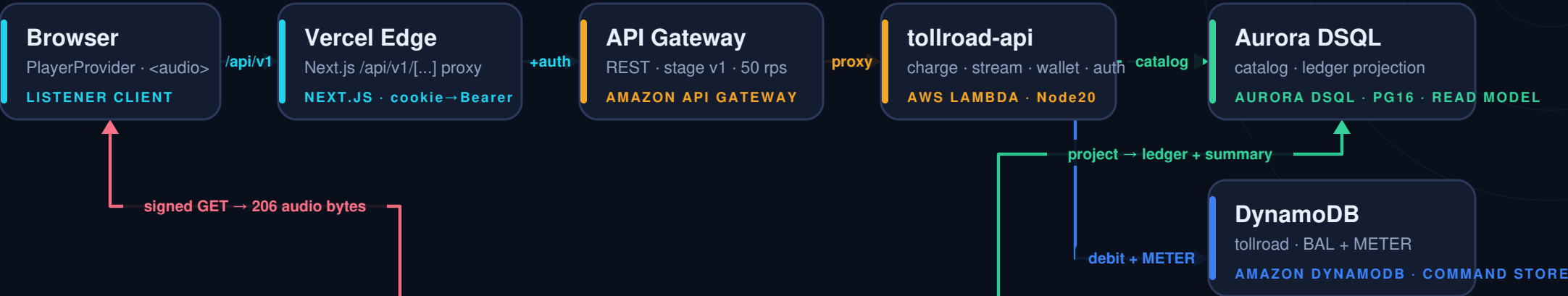


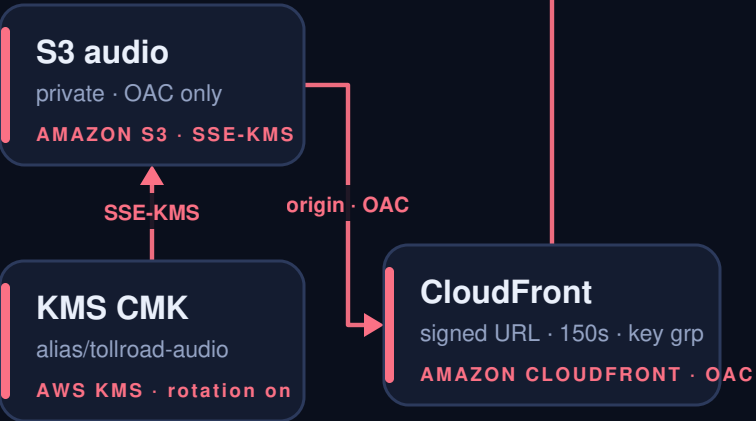
The whole listening machine, one picture

Press play → charge → stream → meter → settle — every component and AWS service

SYNCHRONOUS REQUEST / BILLING PLANE



AUDIO DELIVERY PLANE



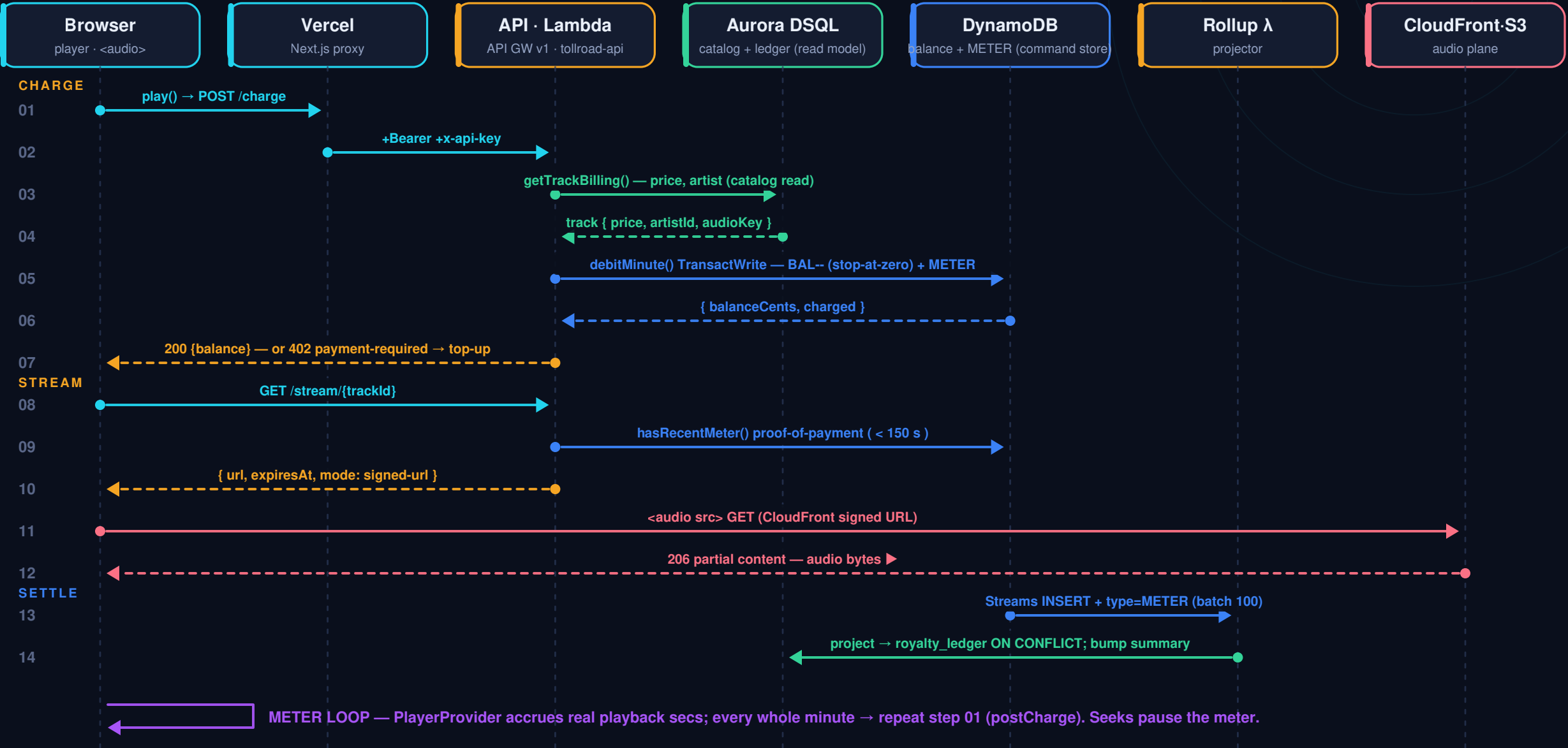
- Client / Vercel
- Compute (API GW · Lambda)
- Aurora DSQL
- DynamoDB
- Audio (S3-KMS-CloudFront)

CQRS: the charge is one DynamoDB transaction (write model); Aurora DSQL is the async projection + catalog. Audio bytes never traverse the API — signed CloudFront serves them edge-direct.

SEQUENCE

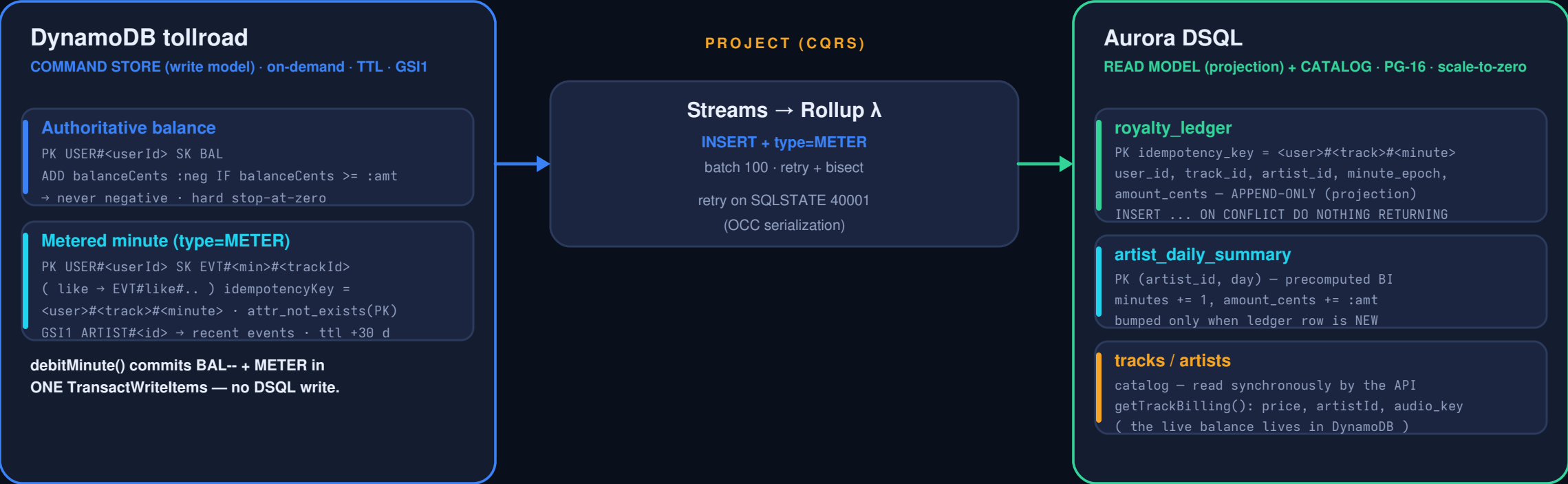
Play → charge → stream → meter → settle

One metered minute, end to end — numbered messages across seven lifelines



Metering — command store, stream, projection

DynamoDB is the command/write store; Aurora DSQL is the projected read model + catalog (CQRS)



EXACTLY-ONCE BY CONSTRUCTION

The idempotency key `<user>#<track>#<minute>` is the same anchor in the DSQL charge txn, the DynamoDB METER item, and the rollup ledger row.

- ① Command store (DynamoDB): BAL debit + METER event commit in ONE TransactWriteItems — balance is authoritative, stop-at-zero.
- ② The METER guard (attribute_not_exists) is checked first → a replayed minute is an idempotent no-op even if the balance moved.
- ③ Stream is at-least-once → projector re-inserts the SAME key → royalty_ledger ON CONFLICT DO NOTHING → exactly-once ledger.
- ④ artist_daily_summary is bumped only when the ledger row is new → artist earnings can never double-count.

Every AWS service in the path

What each construct is, what it does in the listening flow, and how it's configured

SERVICE	CONSTRUCT / NAME	ROLE IN THE LISTENING FLOW	KEY CONFIG
API Gateway	tollroad-api (REST, v1)	Edge entry; API-key usage plan gates the app	50 rps / 100 burst · CORS · webhook key-exempt
Lambda	tollroad-api	All HTTP handlers: charge / stream / wallet / auth / library	Node 20 · 256 MB · 30 s · ESM · proxy integ
Lambda	RollupConsumerFn	Stream projector → builds DSQL ledger + summary (read model)	batch 100 · retry+bisect · pg + dsql-signer layer
Aurora DSQL	cluster (PG-16)	Catalog (sync read) + royalty ledger & summaries (projection)	scale-to-zero · IAM DbConnectAdmin · no FK/triggers
DynamoDB	tollroad (single table)	Command store: authoritative balance (BAL) + METER/TOPUP events	on-demand · TTL 30 d · GSI1 · NEW_AND_OLD_IMAGES
S3	tollroad-audio-<acct>	Encrypted audio objects at rest (private origin)	SSE-KMS · OAC-only read · CORS for Vercel origin
KMS	alias/tollroad-audio	CMK encrypting audio; CloudFront OAC decrypts	rotation on · OAC decrypt policy scoped to distro
CloudFront	TollroadCdn	Serves audio edge-direct via signed URLs (the meter gate)	OAC origin · signed URL / key group · 150 s TTL
S3 + CloudFront	images-<acct> / ImagesCdn	Public cover art delivery	S3-managed enc · OAC · presigned PUT upload
Stripe	PaymentIntent + webhook	Wallet top-ups credit balance idempotently	live mode · /stripe/webhook exempt from API key
Vercel	Next.js app + /api proxy	Player UI + same-origin proxy to the API	httpOnly cookie → Bearer · injects x-api-key

⚠️ DEPLOY DRIFT WATCH — secrets/config a plain `cdk deploy` silently drops or reverts

TOLLROAD_CF_PRIVATE_KEY + TOLLROAD_CF_KEY_PAIR_ID (set outside CDK) · TOLLROAD_SESSION_SECRET · TOLLROAD_SMTP_PASS · all 3 Stripe keys must share mode
Redeploy passing secrets via `-c / --env-file backend/.env` (see safe-backend-deploy); audio streaming requires the CF key pair restored.